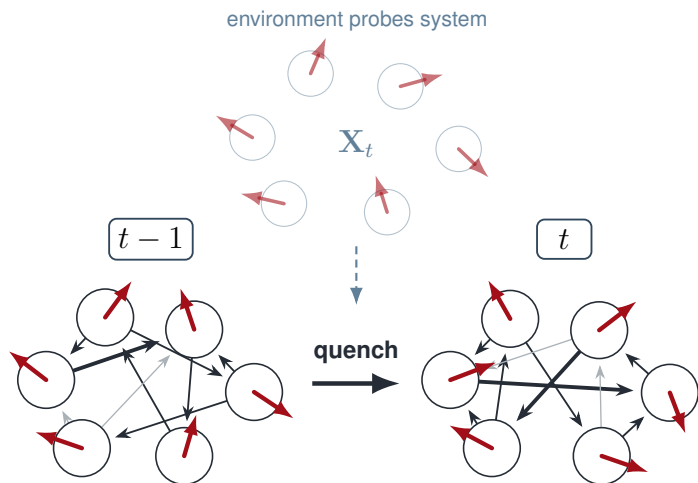


Nonequilibrium dynamics in spin-model transformers

1 Drive quench



The drive quench induces new dynamics

Local field contributions

$$\mathbf{x}_{i,t-1} \longrightarrow \mathbf{x}_{i,t}$$

residual

$$f_{\theta_{\text{FFN}}}(\mathbf{x}_{i,t-1}) \longrightarrow f_{\theta_{\text{FFN}}}(\mathbf{x}_{i,t})$$

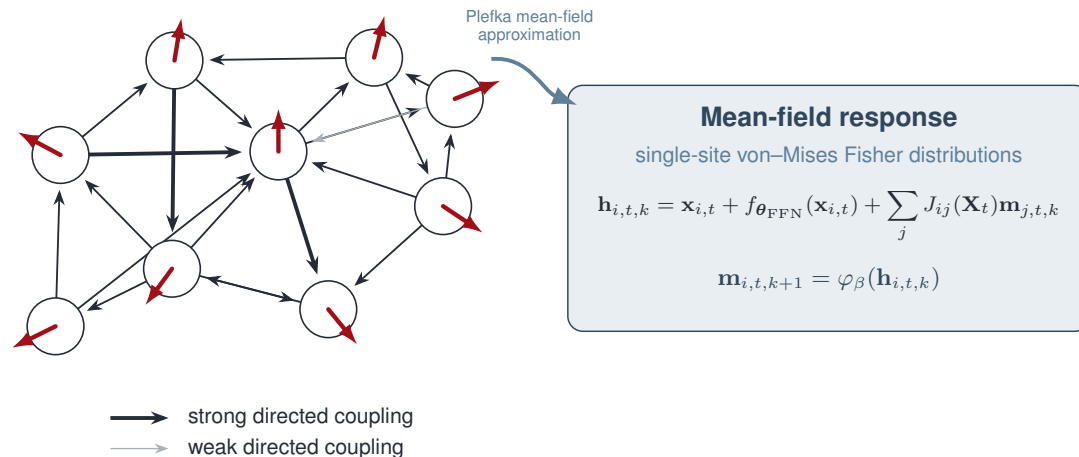
parameterized drive-conditioned field

Directed interactions

$$\mathbf{J}(\mathbf{X}_{t-1}) \longrightarrow \mathbf{J}(\mathbf{X}_t)$$

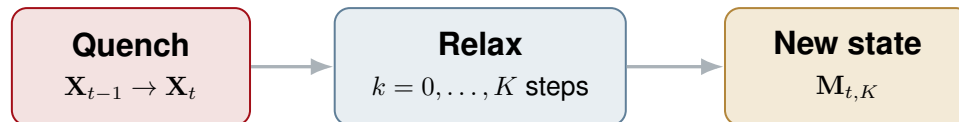
parameterized drive-conditioned couplings

2 Driven disordered vector-spin system



Internal quench-and-relax dynamics

(frozen drive and frozen parameters during relaxation)



3 Design space

